

Jinpeng Lu

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Video World Models | Generative Video Evaluation | Embodied AI



📌 Research Profile & Core Strengths

My research focuses on **video world models** and **generative video evaluation**, particularly world-state persistence under viewpoint changes. I also study **embodied AI** and **vision-language-action (VLA) models**, with an exploratory interest in the verifiability, safety, and evaluation of **recursive self-improvement (RSI)**.

First-author work includes WRBench, an interventional benchmark for persistent world state, and VeloxSeg / H2ASeg for efficient 3D medical perception. I also co-authored VLA-Trace and Pelican-Unify 1.0.

My methods span diagnostic benchmark design, representation and behavior analysis, efficient 3D modeling, and reproducible PyTorch / DDP / CUDA training, evaluation, and ablation pipelines.

🎓 Education

University of Science and Technology of China

Sep. 2025 – Present

M.S. in Information and Communication Engineering, expected Jun. 2028

Advisor: Zhiwei Xiong; interests: video world models, generative video evaluation, embodied AI / VLA models; exploratory interest in verifiability, safety, and evaluation of RSI.

Harbin Institute of Technology, Shenzhen

Sep. 2021 – Jun. 2025

B.S. in Data Science and Big Data Technology (Mathematics discipline); GPA: 3.728/4, Rank: 17/172, Top 10%

Core training: mathematics, statistics, numerical computing, data engineering; research training: PET/CT 3D segmentation, generative signal enhancement.

🏆 Representative Works

Video World Models & Generative Video Evaluation

Current World Models Lack a Persistent State Core

arXiv preprint, First Author

2026.06

- **Question & Approach:** Treated camera motion as an observational intervention and tested whether one task-relevant event state remains coherent through visible evolution, temporary non-observation, and re-observation; built **WRBench** to unify native camera-control interfaces.
- **Evidence & Result:** Diagnosed **23 models** across **four view-synthesis paradigms** and **9,600 videos**, calibrating open-weight evaluators with **4,302 human judgments over 1,741 video pairs**; stronger view synthesis and re-observation support did not reliably yield correct event endpoints.

Embodied Intelligence & VLA

VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing

arXiv preprint, Co-author

2026.05

- **Question & Approach:** Combined representation alignment and drift analysis, causal attention interventions, and behavioral probes to trace VLA models from internal states to decisions and embodied rollouts.
- **Finding:** Experiments on $\pi_{0.5}$ and OpenVLA identified distinct adaptation and action-routing patterns while revealing limits in fine-grained semantic following.

Medical AI

VeloxSeg: JL-Guided Efficient 3D Medical Segmentation

ICLR 2026, First Author [CCF-A]

2026.04

- **Method:** Designed a lightweight CNN-Transformer with multi-scale cross-modal context, JL-guided local 3D features, and training-time texture-prior transfer.
- **Result:** Reached **62.51 / 56.48 Dice** on AutoPET-II / Hecker2022 with **1.66M parameters**, improving GPU/CPU efficiency and memory use.

H2ASeg: Hierarchical PET/CT Tumor Segmentation

MICCAI 2024, First Author [CCF-B]

2024.10

- **Method:** Framed 3D PET/CT tumor segmentation as multimodal volumetric perception and built hierarchical local/global cross-modal interaction with target-aware feature reweighting.
- **Result:** Reached **60.03 / 59.69 Dice** on AutoPET-II / Hecker2022; best-baseline gains: **+2.70 pp / +4.9 pp**.

⚙️ Core Methods & Engineering

Research: Video world models, generative video evaluation, world-state persistence, embodied AI and VLA analysis; medical AI, PET/CT and MRI 3D segmentation

Engineering: Python, PyTorch, CUDA, DDP, Shell/Bash, LaTeX; training/evaluation pipelines, experiment reproduction, ablation studies, custom model implementation

Tools: Linux, Git, Docker, Weights & Biases / TensorBoard, vision, multimodal, and medical-imaging data-processing toolchains

Writing: English manuscript writing for ICLR / MICCAI / Medical Image Analysis submissions, experiment analysis, and reproducibility documentation

■ Full Research Outputs

Video World Models & Generative Video

[1] Current World Models Lack a Persistent State Core, *First Author*

2026 – Present
arXiv 2026

Embodied AI & VLA

[1] VLA-Trace: Diagnosing Vision-Language-Action Models through Representation and Behavior Tracing, *Co-author*

2026 – Present
arXiv 2026

[2] Pelican-Unify 1.0: A Unified Embodied Intelligence Model, *Co-author*

arXiv 2026

Medical AI

[1] Johnson-Lindenstrauss Lemma Guided Network for Efficient 3D Medical Segmentation, *First Author*

2023 – 2026
ICLR 2026 [CCF-A]

[2] H2ASeg: Hierarchical Adaptive Interaction and Weighting Network for Tumor Segmentation in PET/CT Images, *First Author*

MICCAI 2024 [CCF-B]

[3] Does DINOv3 Set a New Medical Vision Standard?, *Co-first Author*

arXiv 2025

[4] AttriMIL: Revisiting Attention-based Multiple Instance Learning for WSI Classification, *Co-author*

Medical Image Analysis
2025 [CCF-C] [SCI Q1]

[5] IPGPhormer: Interpretable Pathology Graph-Transformer for Survival Analysis, *Co-author*

arXiv 2025

[6] A Localization-to-Segmentation Framework for Automatic Tumor Segmentation in Whole-Body PET/CT Images, *Co-author*

arXiv 2023

♀ Additional Project

Generative Sensor Signal Enhancement

Jun. 2023 – Jun. 2024

Lead, National Undergraduate Innovation Program

- Reproduced and compared GAN, diffusion, and flow-based models; designed conditional guidance strategies to improve reconstruction quality and perceptual fidelity, producing a reproducible comparison pipeline and experiment report.